Vehicle Tracking + Occlusions Research Project

Phase 1I Research Documentation

Failure Taxonomy and Systematic Error Analysis

Purpose: archival documentation of the final Phase 1I workflow, manual review decisions, failure taxonomy, final counts, scientific interpretation, limitations, and freeze boundary.

Phase status:	Complete and ready to freeze
Evaluation unit:	12 frozen pilot events × 2 trackers = 24 tracker/event rows
Trackers:	SAM2.1 and DAM4SAM
Verified return events:	10
Verified no-return events:	1 (E012)
Deliberately unresolved:	1 event (E010)
Key output:	Primary + secondary F1–F8 failure taxonomy
Research consequence:	Identity-stage failures, especially F5/F7, justify later sequential vehicle-identity evidence acquisition

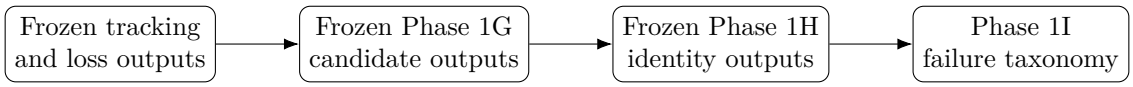
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1 Phase Objective and Research Boundary

Phase 1I was designed to answer a question that the earlier Phase 1 experiments could not answer by themselves: *when the current tracking-and-recovery pipeline fails, what kind of failure actually occurred?*

The purpose is not to improve performance by tuning another algorithm. Instead, Phase 1I converts the frozen baseline behavior into a structured failure taxonomy. This is important because later methodology should be driven by observed failure modes rather than by an assumed novelty.



Frozen finding. Phase 1I is diagnostic, not an optimization phase. Its job is to identify where failures occur so that the next research stage addresses the actual bottleneck.

2 Failure Taxonomy Definitions

The requested taxonomy contains eight failure categories. The final implementation uses a **primary failure** plus optional **secondary failures**; categories are therefore non-exclusive.

Code	Definition
F1	Tracker drifted before occlusion.
F2	Target became completely lost during the evaluated episode.
F3	Tracker jumped to a similar distractor vehicle.
F4	True target returned but was not detected / retained by the candidate generator.
F5	True target was detected but the identity matcher did not select it Rank-1.
F6	Target identity memory became contaminated by incorrect target observations.
F7	Multiple candidates remained genuinely ambiguous after return.
F8	Target never returned during the evaluated observation window.

Two interpretation rules are important. First, F8 is an event condition rather than an identity-selection error: if the target never returns, the recovery system cannot be expected to identify it. Second, F3 is tracker-specific, while F5/F7 are primarily identity-stage event findings and therefore appear in both tracker rows for the same event.

3 Evidence Base and Analysis Contract

Phase 1I uses the same 12 frozen UA-DETRAC pilot events E001–E012 and the two real tracking baselines SAM2.1 and DAM4SAM. The master taxonomy therefore contains 24 tracker/event rows.

Evidence is drawn only from already frozen Phase 1 artifacts:

- Phase 0 manual event review and frozen event log.
- Phase 1E event-aware loss / recovery decisions.
- Phase 1F L1/L2 loss-state diagnostics.
- Phase 1G real YOLO11n candidate-generation results.
- Phase 1H H0/M1/M2 identity-ranking, memory, ambiguity, and reliability diagnostics.
- Phase 1I original-frame manual review for F3 and F7 cases.

Conservative rule. XML annotation presence is not treated as proof that a target is visually visible during occlusion. Original-frame review remains authoritative for visibility and distractor-jump decisions.

4 Phase 1I Workflow

The analysis was deliberately staged so that labels were not assigned before the underlying facts were established.

1. Create a 24-row taxonomy skeleton with all failure fields initially unknown.
2. Populate objective return / detection / identity facts from frozen Phase 1G/1H results.
3. Populate genuine-loss facts from Phase 1E.
4. Mark suspected F3 cases and manually review original frames.
5. Diagnose F1 using pre-occlusion tracker-vs-GT overlap, followed by conservative interpretation of non-assessable initialization cases.
6. Identify F7 suspects from identity-stage ambiguity diagnostics and manually compare the true target against the highest-ranked wrong candidates.
7. Assign final primary and secondary F1–F8 labels using a fixed hierarchy.
8. Produce tracker-row and unique-event failure summaries.

The final hierarchy used for primary labels was:

- F8 primary when the target never returned.
- F7 primary when genuine ambiguity was manually confirmed and the target was detected but not selected.
- F5 primary when the target was detected but identity selection failed without confirmed F7.
- F4 primary when a returned target was not retained by the candidate generator.
- F2 primary when genuine target loss occurred but later candidate/identity stages succeeded.
- F3, F6, and F1 become primary only when they are the dominant remaining failure; otherwise they can appear as secondary evidence.

5 Objective Return, Detection, and Identity Facts

Before assigning failure labels, Phase 1I populated facts already established by earlier phases.

Fact	Frozen Phase 1 evidence
Verified target return	10/12 events.
Verified no return	E012.
Unresolved return status	E010; deliberately not forced.
Phase 1G target retention	10/10 eligible verified-return events.
M1 Rank-1 identity success	5/10 eligible events.
M1 Rank-1 identity failure	5/10 eligible events: E001, E003, E004, E008, E011.
M1 memory contamination	0 contaminated entries in the 26-entry memory-quality audit.

This already constrains several failure categories: F4 is absent from the 10 eligible verified-return events, while F5 is clearly present in five events.

6 F1 - Pre-Occlusion Drift Analysis

F1 was tested using tracker-vs-GT IoU from the selection frame up to, but not including, the recorded occlusion start. A diagnostic threshold of IoU < 0.30 was used only to identify manual-review suspects; it was not a final system parameter.

The result was strong: no assessable tracker/event run contained a sustained pre-occlusion low-overlap sequence.

Outcome	Result
F1 confirmed YES	0/24 tracker rows.
F1 confirmed NO	16/24 tracker rows.
F1 not assessable	8/24 rows, corresponding to E001, E002, E004, and E012 for both trackers because occlusion begins essentially immediately after initialization.
Manual-review suspects	0/24.

For assessable cases, the pre-occlusion mean IoU remained high (for example, approximately 0.91 in E003, 0.77–0.78 in E005, 0.75–0.77 in E006, 0.95–0.96 in E008, and 0.84–0.85 in E009).

Phase 1I conclusion for F1. There is no confirmed evidence that pre-occlusion drift is a dominant failure in this pilot.

7 F2 - Genuine Target Loss

Phase 1E had already separated continuous tracking from genuine loss. Phase 1I carries those frozen decisions into the taxonomy.

Status	Events
Verified genuine loss	E005, E006, E007, E012.
No verified genuine loss	E001, E002, E003, E004, E008, E009, E011.
Unresolved	E010.

Thus F2 occurs in 4/12 unique events. Because E012 has no verified reappearance, F8 is primary there and F2 is retained as secondary evidence.

8 F3 - Jump to Similar Distractor

Two SAM2 runs were marked as suspected F3 cases from earlier tracking review: E006/SAM2 and E012/SAM2. Both were manually reviewed on original frames with SAM2 predictions, target annotations, and other vehicle annotations overlaid.

8.1 E006 / SAM2

In E006, SAM2 loses output around frames 1094–1100 and later resumes on a different nearby vehicle while the target remains elsewhere. This is a confirmed tracker jump to a distractor.



Figure 1: E006 manual F3 review. Red = SAM2 prediction, green = target annotation, yellow = other annotated vehicles. The later SAM2 box resumes on a different vehicle.

8.2 E012 / SAM2

In E012, after the target is no longer present in the evaluated observation window, SAM2 continues producing a box on another visible vehicle. This is also recorded as F3, but only as a secondary failure because F8 is the dominant event condition.



Figure 2: E012 manual F3 review. SAM2 continues on another visible vehicle after the target is no longer verified to return.

F3 result. Confirmed F3 occurs in 2/12 unique events (E006 and E012), and only in the SAM2 tracker rows. DAM4SAM did not receive an F3 label in these manually reviewed cases.

9 F4, F5, and F6 - Candidate, Identity, and Memory Failures

9.1 F4 - Returned target not detected

Phase 1G retained the true target candidate in all 10 eligible verified-return events. Therefore no F4 event was observed in this pilot.

9.2 F5 - Target detected but not selected

F5 is the most frequent recoverable identity-stage failure. The true target was present in the Phase 1G candidate set but M1 failed to rank it first in:

E001, E003, E004, E008, and E011.

This is 5/12 unique pilot events, or 5/10 of the events for which verified reappearance identity ranking was evaluated.

9.3 F6 - Memory contamination

The corrected M1 pre-occlusion memory audit checked 26 recent SAM2 target-memory entries. Twenty-five had strong tracker-vs-GT IoU ≥ 0.50, one was weak (0.30–0.50), and none had IoU < 0.30. No F6 event was therefore confirmed.

10 F7 - Genuine Candidate Ambiguity

F7 required manual evidence that the *true target itself* remained plausibly confusable with high-ranked wrong candidates. A small Top1–Top2 score margin alone was not treated as sufficient proof.

The manual decisions were:

Event	F7	Manual interpretation
E001	Yes	True target and several top wrong candidates are small light/silver vehicles with similar frontal appearance.
E003	No	True target is visibly different in scale/viewpoint from the highest-ranked wrong vehicles; better explained as F5 representation/ranking failure.
E008	Yes	True dark vehicle is Rank 4 among several closely similar dark vehicles.
E011	Yes	Multiple taxi-like vehicles share very similar color/livery and side-view appearance.

10.1 Confirmed F7 examples

10.2 Contrast case: E003

F7 result. Genuine candidate ambiguity is confirmed in 3/12 unique pilot events: E001, E008, and E011. In each of these events, F5 also holds because the correct target was present but was not selected Rank-1.



Figure 3: Confirmed F7 examples. E001 (top), E008 (middle), and E011 (bottom) show the true target competing with visually similar alternatives.



Figure 4: E003 is not labeled F7. The true target is sufficiently different from the top wrong candidates; the failure is retained as F5 rather than genuine ambiguity.

11 Final Primary + Secondary Taxonomy

The final event-level interpretation is shown below. Where tracker behavior differs, the tracker-specific secondary label is stated explicitly.

Event	Primary	Secondary	Interpretation
E001	F7	F5	Genuine ambiguity among visually similar vehicles; target detected but not selected.
E002	NONE	–	No classified failure in the evaluated recovery/identity episode.
E003	F5	–	Target detected but appearance matcher ranked visibly different wrong candidates higher.
E004	F5	–	Confidently wrong identity selection; target detected but ranked poorly.
E005	F2	–	Genuine loss occurred, followed by successful target candidate retention and Rank-1 identity selection.
E006	F2	F3 on SAM2	Genuine loss for both trackers; SAM2 additionally jumped to a distractor.
E007	F2	–	Genuine loss occurred, later recovery/identity succeeded.
E008	F7	F5	Genuine ambiguity among similar dark vehicles; target Rank 4.
E009	NONE	–	No classified failure in the evaluated episode.
E010	UNRESOLVED	–	Target remained occluded/unresolved; return status deliberately not forced.
E011	F7	F5	Genuine taxi-like distractor ambiguity; target detected but not selected.
E012	F8	F2; F3 on SAM2	No verified return. Genuine loss is secondary; SAM2 also jumped to another vehicle.

11.1 Tracker-row final assignments

Event	Tracker	Primary	Secondary
E001	SAM2.1	F7	F5
E001	DAM4SAM	F7	F5
E002	SAM2.1	NONE	–
E002	DAM4SAM	NONE	–
E003	SAM2.1	F5	–
E003	DAM4SAM	F5	–
E004	SAM2.1	F5	–
E004	DAM4SAM	F5	–
E005	SAM2.1	F2	–
E005	DAM4SAM	F2	–
E006	SAM2.1	F2	F3
E006	DAM4SAM	F2	–
E007	SAM2.1	F2	–
E007	DAM4SAM	F2	–
E008	SAM2.1	F7	F5
E008	DAM4SAM	F7	F5
E009	SAM2.1	NONE	–
E009	DAM4SAM	NONE	–
E010	SAM2.1	UNRESOLVED	–
E010	DAM4SAM	UNRESOLVED	–
E011	SAM2.1	F7	F5
E011	DAM4SAM	F7	F5
E012	SAM2.1	F8	F2, F3
E012	DAM4SAM	F8	F2

12 Consolidated Failure Counts

Two different count views are required. The **unique-event view** is the scientifically meaningful view for event-level F5/F7 findings. The **tracker-row view** preserves tracker-specific behavior such as F3.

12.1 Unique events containing each failure

Code	Failure	Count	Events
F1	Pre-occlusion drift	0/12	–
F2	Genuine target loss	4/12	E005, E006, E007, E012
F3	Jump to distractor	2/12	E006, E012
F4	Returned but not detected	0/12	–
F5	Detected but not selected	5/12	E001, E003, E004, E008, E011
F6	Memory contamination	0/12	–
F7	Genuine ambiguity	3/12	E001, E008, E011
F8	No verified return	1/12	E012

12.2 Primary failure counts across 24 tracker rows

Primary label	Rows	Primary label	Rows
F1	0	**F6**	0
F2	6	**F7**	6
F3	0	**F8**	2
F4	0	**NONE**	4
F5	4	**UNRESOLVED**	2

The zero primary count for F3 does not mean F3 never happened. It was confirmed in E006/SAM2 and E012/SAM2, but in both cases a more fundamental event condition (F2 or F8) was primary.

13 Research Interpretation

Phase 1I produces a clear diagnosis of the current pipeline.

1. **Pre-occlusion drift is not the dominant bottleneck.** No F1 case was confirmed in the assessable pilot runs.
2. **Candidate generation is not currently the main bottleneck.** No F4 case occurred in the 10 verified-return identity events; Phase 1G retained the target in all of them.
3. **Identity selection is the dominant recoverable failure.** F5 occurs in five unique events, more than any other recoverable identity-stage category.
4. **A substantial subset of identity failures is genuine ambiguity.** Three of the five F5 events also contain confirmed F7 ambiguity.
5. **Tracker loss still matters.** F2 occurs in four events, and F3 occurs in two SAM2 runs, but these do not explain the majority of verified-return identity failures.
6. **Memory contamination was not observed in the corrected M1 audit.** This weakens the case for treating contamination as the immediate next design priority.

Central Phase 1I finding. In the current pilot, the main recoverable bottleneck is not simply finding a returned vehicle. The system frequently contains the correct vehicle in its candidate set but cannot reliably distinguish it from plausible alternatives using appearance alone.

This directly supports the planned next research stage: reliability-aware sequential acquisition of additional vehicle-identity evidence rather than always-on multimodal computation or forced appearance-only decisions.

14 Scientific Cautions and Limitations

The following cautions are frozen with the Phase 1I interpretation:

- The pilot contains only 12 manually curated events and 10 verified-return identity cases. Counts guide methodology; they are not final population estimates.
- F5/F7 event findings are duplicated across SAM2.1 and DAM4SAM rows because the same Phase 1G/1H identity evaluation was used; unique-event counts must therefore be reported alongside tracker-row counts.
- F7 required manual visual confirmation. Low score margin alone was not used as proof of genuine ambiguity.
- E010 remains unresolved by design. The taxonomy does not force F8 or another label where return/absence is not verified.
- E001, E002, E004, and E012 are not assessable for F1 because the occlusion interval begins essentially immediately after initialization.
- F8 is a condition of the observation window, not an algorithmic identity error.
- Thresholds previously used for Phase 1H selective routing were pilot-derived and are not treated here as validated taxonomy rules.

15 Implementation Artifacts

The Phase 1I record is represented by the following scripts and outputs:

- `src/evaluation/build_phase1i_taxonomy_skeleton.py`
- `src/evaluation/populate_phase1i_objective_facts.py`
- `src/evaluation/populate_phase1i_tracking_facts.py`
- `src/evaluation/visualize_phase1i_f3_suspects.py`
- `src/evaluation/confirm_phase1i_f3_manual_review.py`
- `src/evaluation/diagnose_phase1i_f1_preocclusion_drift.py`
- `src/evaluation/confirm_phase1i_f1_results.py`
- `src/evaluation/visualize_phase1i_f7_suspects.py`
- `src/evaluation/visualize_phase1i_f7_target_vs_top.py`
- `src/evaluation/confirm_phase1i_f7_results.py`
- `src/evaluation/finalize_phase1i_failure_taxonomy.py`
- `results/phase1/phase1i_failure_taxonomy_final.csv`
- `results/phase1/phase1i_failure_summary.csv`

16 Phase 1I Freeze Decision and Handoff

Phase 1I is considered complete when the final taxonomy and summary are accepted. No additional threshold tuning or tracker modification is part of this phase.

Frozen decision	Status
F1	No confirmed pre-occlusion drift; 8 rows not assessable.
F2	Confirmed in E005, E006, E007, E012.
F3	Confirmed in E006/SAM2 and E012/SAM2.
F4	No observed case among 10 eligible verified-return events.
F5	Confirmed in E001, E003, E004, E008, E011.
F6	No confirmed contamination in corrected M1 audit.
F7	Confirmed in E001, E008, E011; rejected for E003.
F8	Confirmed for E012.
E010	Remains unresolved.

Freeze statement. Phase 1I establishes the final Phase 1 failure taxonomy. The evidence points most strongly toward post-return identity selection and genuine look-alike ambiguity, providing the empirical justification for moving from baseline tracking/recovery analysis to vehicle-identity evidence and sequential acquisition in the next research stage.

End of Phase 1I Research Documentation